Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/16/2024
PI Name:	Matson, Douglas Ph.D.		
Project Title:	PSI Revisiting EML Video Images from EML Work (PREVIEW)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-PSI E.8 Physical Sciences Informatics
Start Date:	12/15/2023	**End Date:**	12/14/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nawer, Jannatun Ph.D. (Tufts University)		
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Task Description:	This study will leverage the existing dataset from the NASA Physical Sciences Informatics (PSI) database to re-evaluate thermophysical property data from the TEMPUS electromagnetic levitator (EML) aboard the Space Shuttle Columbia. Lessons learned from the current International Space Station (ISS) EML programs can be applied to previous microgravity datasets generated during the Space Shuttle era. From this, we hope to extract thermophysical properties with both higher accuracy and greater precision. [Ed. Note: TEMPUS is a German acronym (Tiegelfreies Elektromagnetisches Prozessieren unter Schwerelosigkeit).] Aims - The key scientific questions that will be addressed during this study are: 1. Re-evaluate old data based on insights gained during subsequent years of investigation. 2. Study and report accuracy and precision during thermophysical property measurement comparing previous microgravity experiments to those currently ongoing on the ISS. Objectives – The work is structured to deliver the following tasks: 1. Re-evaluate surface tension based on new information on the impact of sample deformation. 2. Re-evaluate viscosity results based on new information on the impact of pulse induced turbulence and using new magnetohydrodynamic (MHD) surrogate modelling. 3. Conduct uncertainty analysis using GUM methodologies. [Ed. Note: "GUM" refers to the ISO Guide to the expression of Uncertainty Measurement.] Approach - To satisfy the objectives: 1. A correction factor will be applied to Rayleigh's formula to account for the non-linear behavior observed with a larger deformation amplitude. 2. Viscosity will be corrected, based on identifying observed flow regimes on the overshoot decay pattern at higher Reynolds number. 3. MHD surrogate model will be applied to identify the effect of static field induced convection on the viscosity measurement. 4. Uncertainty analysis will be performed using GUM methodologies. Significance - If the aims of the proposal are achieved, the following outcomes are expected: 1. Advance the understanding of how the thermophysical data is influenced by previously unknown measurement biases. In the future, this will help the researchers to conduct better experiments to leverage the unique environment of microgravity. 2. Initiate activity across the property measurement community to encourage standards reporting of uncertainty of the measurements to help the scientific and industrial design communities to better understand error propagation and to increase the fidelity of risk assessment for design of aerospace hardware supporting human exploration of space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Impact: 1) This research will advance the understanding of how the thermophysical data is influenced by previously unknown measurement biases. 2) Initiate activity across the scientific community to encourage standardized uncertainty reporting.
Task Progress:	1. Data Procurement: A thorough search for Shuttle experiments data and to-date was initialized. Several videotapes from Principal Investigator (PI) Douglas M. Matson's archive at Tufts University, Medford, MA. Additional datasets were found in the PSI archive and we are currently in the process of downloading the data. 2. Data Conversion: VHS tapes obtained were in European PAL format and a professional service was rendered for the digitization of these video files to MP4 format. 3. Data Analysis: An initial quality check has been performed in the acquired videotapes for determining the feasibility of edge detection for surface tension analysis. Steps have been performed using a custom MATLAB algorithm. Another software, RAUDEC2, was obtained from the PSI database for data extraction and we are currently in the process of understanding how the software works. A custom algorithm is being developed in the Open source software Python to enhance image quality and perform sub-pixel edge detection on the raw data for surface tension and viscosity analysis.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Abstracts for Journals and Proceedings	Nawer J, Matson DM. "Re-evaluating thermophysical property data from Space Shuttle Columbia - TEMPUS EML." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024